1 Progress Summary

I have completed initial validation of morphological classifiers using Euclid Early Release Objects (ERO) and Quarter 1 (Q1) data cross-matched with Gaia DR3. Key progress includes:

- Cross-matched Q1 sources with Gaia DR3 using `Gaia_id` from Q1 merged catalogue
- Identified systematic magnitude-dependent bias rendering `spread_model` unreliable
- Determined optimal matching radius (0.2 arcsec) through Rayleigh distribution analysis
- Cross-matched ERO sources with Gaia DR3 using position-based matching (`m.v_alpha_j2000`, `m.v_delta_j2000`)

2 Preliminary Findings

2.1 Euclid ERO Classifiers: `class_star` vs `spread_model`

`class_star` is strongly preferred over `spread_model` for extended source identification. Analysis reveals critical magnitude-dependent systematic bias in `spread_model` rendering it fundamentally unreliable across all magnitude ranges.

Both Euclid ERO classifiers were validated across $10 < G < 27$ mag. The `spread_model` parameter exhibits a characteristic two-bump structure (Figure 1) caused by magnitude-dependent effects:

- **Bright regime** ($G < 15$ mag): High SNR measurements produce tight astrometric fits, causing genuinely extended sources with small spread values to be misclassified as point-like (first bump at $G \sim 12$–15).
- **Faint regime** ($G > 20$ mag): Low SNR produces inflated spread values from measurement noise, causing point-like sources to be misclassified as extended (second bump at $G \sim 20$–22).
- **Intermediate regime** ($15 < G < 18$ mag): Temporarily reasonable classification, but this narrow range is insufficient for general-purpose use.

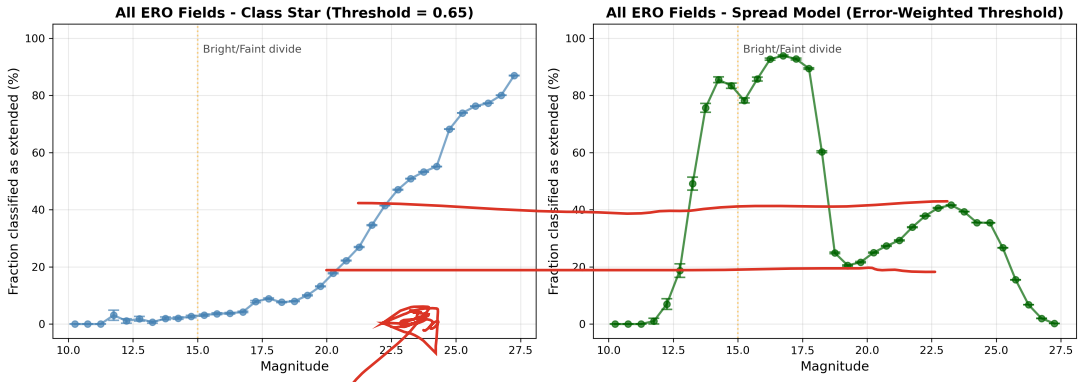


Figure 1: Sanity checks for Euclid ERO classifiers. Left: `class_star` shows expected monotonic increase in extended fraction with magnitude. Right: `spread_model` exhibits two-bump structure from SNR-dependent systematic bias. Error bars show binomial uncertainties.

Between roughly 20 and 22.5 both methods are giving the same results - yet in your description you say that class star is correct, but that spread model is wrong in that regime - both statements cannot both be true - you might want to have another think about this.
If class star is correct this implies that spread model has an excess of extended sources at brighter magnitudes and is missing extended sources at fainter magnitudes.

You might want to think about this again

Figure below demonstrates the SNR-`spread_model` correlation. High-SNR sources cluster around `spread_model` $\sim$ 0 even for genuinely extended objects, while low-SNR sources show broad distributions with increased scatter, confirming the parameter's unreliability.

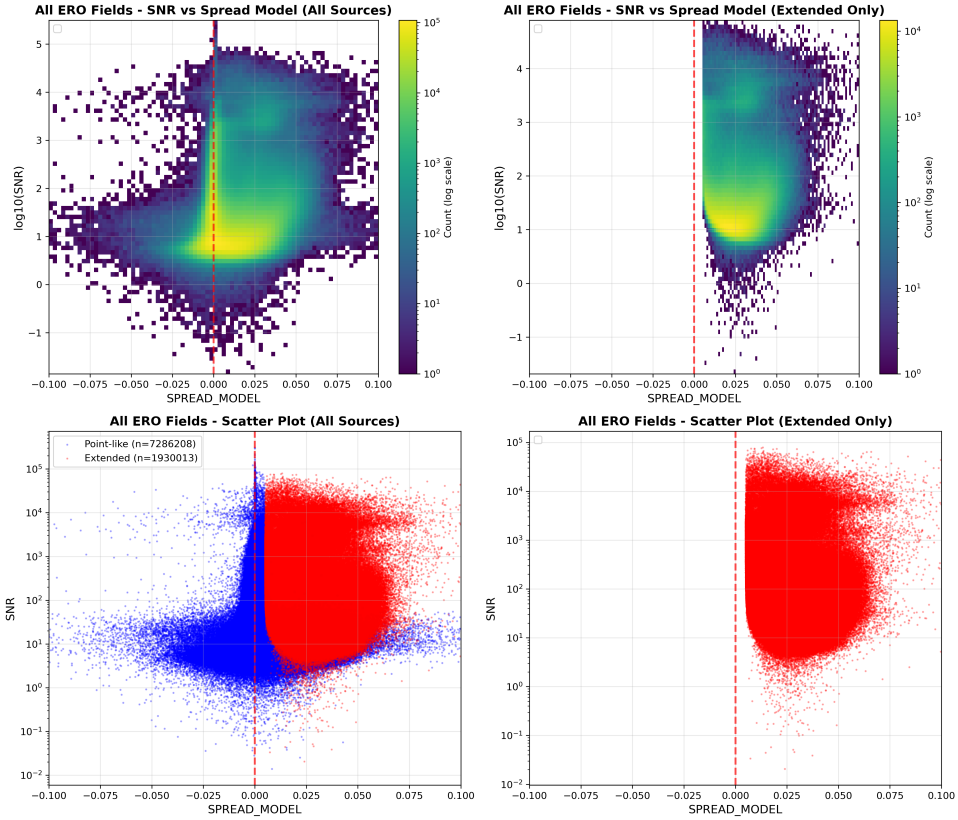


Figure 2: SNR vs `spread_model` for ERO sources. Top: 2D density plots (log scale) for all sources (left) and extended sources only (right). Bottom: Scatter plots with point-like (blue) and extended (red) sources. Strong SNR dependence confirms systematic bias.

2.2 Q1 Photometric Classification Validation

Euclid Q1 includes PHZ pipeline classifications (STAR, GALAXY, QSO, OTHER).

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3 Cross-Matching Methodology

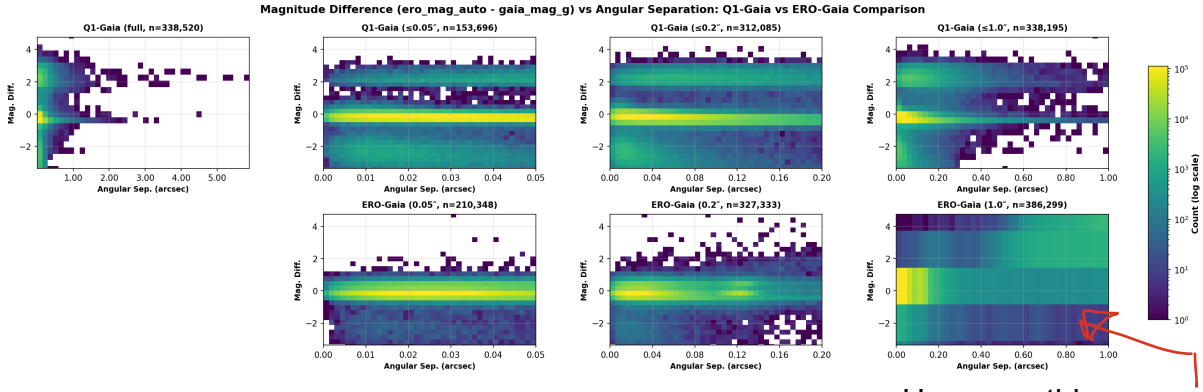
3.1 Photometric Consistency Check

To understand ERO-Gaia cross-matching behavior, photometric consistency across multiple matching radii (0.05, 0.2, 1.0 arcsec) was first examined. Figure 3 shows magnitude differences (Euclid − Gaia) remain consistent across radii for both Q1-Gaia and ERO-Gaia cross-matches. Tight radii (0.05 arcsec) select secure matches with minimal scatter; looser radii (0.2, 1.0 arcsec) include more matches with slightly increased scatter but no systematic offset.

This exploratory analysis indicated that truncation around 0.2 arcsec would effectively separate genuine matches from spurious associations, informing the choice of truncation radius for

I think for this plot you really want to use the same magnitude for both Q1 and ERO as otherwise we don't really learn much from the comparison

subsequent statistical fitting.



Has something gone wrong with the binning here?

Figure 3: Photometric consistency at multiple radii. Magnitude difference vs angular separation for Q1-Gaia (top) and ERO-Gaia (bottom) at 0.05, 0.2, 1.0 arcsec. Color shows source density (log scale).

3.2 Optimal Matching Radius Determination

Building on the photometric consistency analysis, statistical determination of the optimal matching radius was performed. Angular separations between matched pairs are expected to follow Rayleigh distributions .

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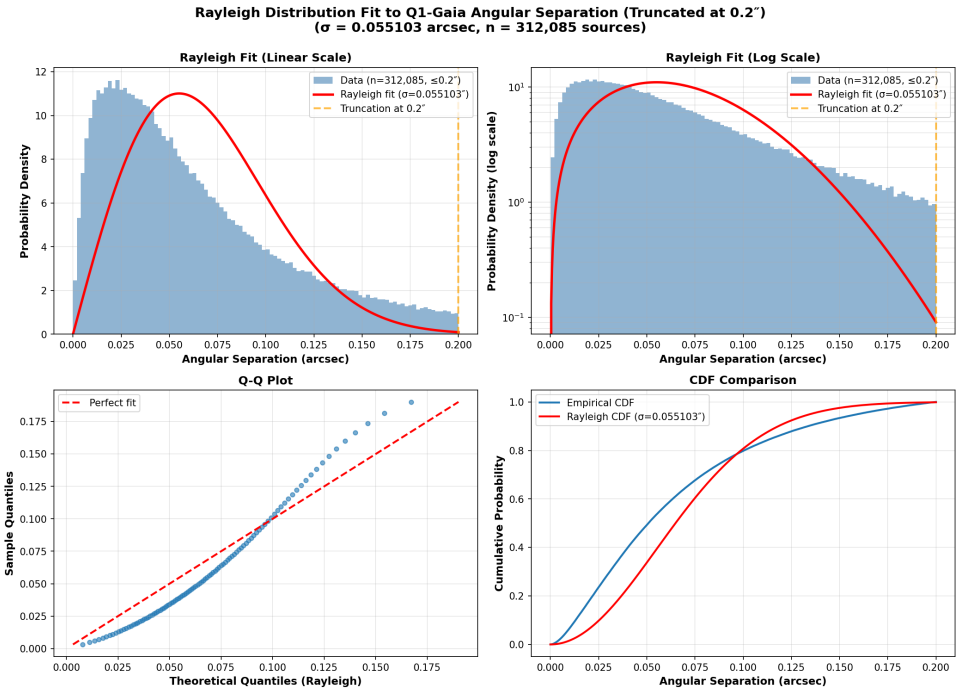


Figure 4: **Rayleigh fit to Q1-Gaia angular separations.** Blue histogram with fitted Rayleigh PDF ($\sigma = 0.0920$ arcsec, red curve). Panels: (top-left) linear scale; (top-right) log scale; (bottom-left) Q-Q plot; (bottom-right) CDF comparison. Excellent fit validates 0.2 arcsec matching radius.

This is not an excellent fit. Are you using the errors ?

4 Planned Analyses and Timeline

4.1 Upcoming Analyses

1. **PSF flux to magnitude calibration:** Understand conversion from PSF flux measurements to magnitudes and identify systematic biases in PSF-based photometry for extended sources
2. **Magnitude differences vs magnitude:** Plot Δmag (Euclid − Gaia) vs magnitude to identify magnitude-dependent systematic offsets
3. **Magnitude differences vs color:** Examine Δmag vs BP−RP (Gaia color) to identify color-dependent photometric systematics
4. **Magnitude differences by morphological class:** Compare Δmag vs magnitude for extended vs point-like sources separately to quantify classification-dependent photometric biases
5. **Extended sources in color-magnitude space:** Investigate where extended sources are located in BP−RP vs magnitude diagrams relative to point sources
6. **ML feature selection:** Identify optimal Gaia parameters for classification based on validation results
7. **XGBoost training:** Develop initial classifier using validated features with Euclid morphological labels as ground truth

4.2 Timeline

Next 2–3 weeks:

- Complete Euclid classification analysis
- Execute analyses 1–5 above
- Finalize ML feature set based on validation results

Following month:

- Begin XGBoost model development
- Feature importance analysis and engineering iteration
- Initial performance evaluation on test set

I know this is difficult, but I think you need to have in your timeline a plan which takes you all the way to the end of the project.

What do you want to achieve by the end of the project? And how are you going to get there?

So after you learn from XGBoost which Gaia parameters tell you that a source is likely an extended source - what do you want to do then?

Do you want to do some clustering analysis (UMAP / Self-organising maps) to see whether Gaia can distinguish between different kinds/morphologies of extended sources?

There is also the possibility to look at the images from Euclid for sources from different clusters to see what the ground truth is...

It is okay to have a plan which has a "if time allows" section.